



CEO age and risk-taking of family business in Malaysia: The inverse S-curve relationship

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Abstract

The relationship between CEO age and corporate risk-taking has remained inconclusive after decades of research. This study contends that CEO may have different risk-taking propensities at different age stages and the relationship could be better captured by nonlinear function. Using a large sample of 6169 firm-year observations of public listed companies in Malaysia, we study the risk-taking behaviour of CEO for the period of 2009 to 2017. We find that CEO age has an inverse U-shaped relationship, such that risk-taking increases with CEO age but reduces beyond certain age threshold. Besides, drawing on socioemotional wealth perspective, we further examine the moderating effect of family CEO. We find that near-retirement family CEO reverses the conservative behaviour at old age and results an inverse S-curved relationship. Findings of this paper yields implication for family business and policymakers seeking to encourage risk-taking behaviour in emerging countries.

Keywords Nonlinear relationship · CEO age · Risk-taking · Family business

Previous research has suggested that CEOs age is a major factor affecting the risk-taking activities of a firm (Li, Low, & Makhija, 2017; Serfling, 2014; Yim, 2013). Despite decades of established body of empirical investigations, the linkage between risk-taking and CEO age is mixed and inconclusive. Majority of research argue that older CEOs are less willing to take risk due to aging effect (Child, 1974; Hambrick & Mason, 1984; Taylor, 1975) and prefer quiet life when they get older (Bertrand & Schoar, 2003). Firms managed by older CEO have lower leverage ratio (Serfling, 2014) and have less R&D spending (Li et al., 2017). In contrast, some scholars argue that older adults base the decision-making on more experiential strategies rather than

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analytic perspective due to frontal lobe function (Weller, Levin, & Denburg, 2011). As a result, older CEOs are more willing to take risk as they have greater experience in identifying risky strategies that produce greatest chances of success (Simsek, 2007). Probability domain familiarity by Sitkin and Pablo (1992) also suggests that the greater the past experience of a decision maker in dealing with risk, the less likely s/he is to perceive uncertainty on the outcome of taking these risks, and the more reasonable the risk will seem to be. Most of the prior research that study the relationship between CEO age and managerial risk-taking focus on linear relationship as age moves uniformly with time.

While we acknowledge that there are merits to the logics examined in previous studies, we contend that overall relationship between CEO age and firm risk-taking may be better captured by nonlinear relationship. It can be seen from most of the financial portfolio research on individual and household reveal that age of an individual indeed has an inverse U-shaped or hump-shaped with his/her risk tolerance level (Chambers & Schlagenhauf, 2002; Halek & Eisenhauer, 2001; Riley Jr & Chow, 1992). These prior empirical studies find that life cycle stage of an individual has a significant impact on his/her risk-taking behaviour in terms of holding of risky asset. According to life-cycle hypothesis of asset allocation, individual's holding of risky asset such as equities varies at different life cycle due to earnings and accumulated wealth at different age stages (Riley Jr & Chow, 1992).

Based on this notion, we attempt to explain the nonlinear risk-taking behaviour of CEO based on their career life cycle. In the beginning of career life, younger CEOs have higher tendency to undertake more aggressive investment policies to signal their superior ability (Bebchuk & Stole, 1993; Prendergast & Stole, 1996). However, beyond a certain age threshold, the tendency of taking risk is likely to be influenced more significantly by career horizon effect. CEOs are reluctant to engage in risky strategies when nearing retirement due to shortened horizon (McClelland, Barker III, & Oh, 2012). This leads to an inverse U-shaped relationship between CEO age and corporate risk-taking.

Drawing on socioemotional wealth literature, we contend that CEOs in family firms differ in their risk-taking behaviour as the career horizon problem is weaker for family CEO. The near-retirement family CEOs are more willing to engage in risk-taking activities to secure growth of the family business for their future generation (Strike, Berrone, Sapp, & Congiu, 2015). The downward curve of inverse U-shaped relationship is reversed due to weaker career horizon problem and this leads to an inverse S-curve relationship.

We test the possible existence of nonlinear relationship between CEO age and risk taking in the context of Malaysia. There are many old-aged CEOs in Malaysia who still hold the position into their 70s and 80s. They are mostly the founders of the firms who have started the family business since independence of Malaysia in year 1957. Family business group in Malaysia emerged during the period of colonial disengagement and establishment of nationalistic government after the World War II period in the early 20th centuries. They develop their major entrepreneurial forces in 1960s when filling the vacuum in plantation agriculture, mining and oil extraction left by departing colonial powers (Carney & Gedajlovic, 2003). Since then, these family firms begin to accumulate wealth and dominate the organisational forms and corporate structure of Malaysia.

Among East Asian countries, Malaysia has the third highest concentration of control after Thailand and Indonesia by which firms are tightly held by family member (Thillainathan, 1999). Family ownership is a prevalent firm structure in Asian corporations. The largest ten families control half of the corporate assets in Indonesia, Thailand and the Philippines and nearly one-third in Korea and Hong Kong (Claessens, Djankov, & Lang, 2000). The top 15 family-controlled firms account for 84% of gross domestic product (GDP) in Hong Kong, 48% in Singapore, 46% in the Philippines and 39% in Thailand (Morck, Wolfenzon, & Yeung, 2005). For Malaysia, family firms account for nearly 70% of the firms listed in Bursa Malaysia (Afza Amran & Che Ahmad, 2009) and contribute approximately 67% of nominal GDP in Malaysia (Fan, Tan, Guller, Garcia, & Ouek, 2011).

Similar as other family firms in Asia, family firms in Malaysia are characterized by blood ties and based on the notion of family continuity due to family obligation. Family firms are often run by family CEOs than professional CEOs and only family heirs are chosen as succeeding CEOs. Family succession is prevalent in family firms in most of the countries in Asian and emerging countries. Thailand family business follow primogeniture principle of passing the leadership to the first born children to avoid rivalry and friction (Yabushita & Suehiro, 2014); Hong Kong family founders incubate next generation in spin-offs of the family firms (Au, Chiang, Birtch, & Ding, 2013); Founder of firms in China groom their future male heirs by allowing them to serve on the management team and board of directors (Weng & Chi, 2019). In Malaysia, father-entrepreneurs emphasize transgenerational control and dynastic continuity of family business by placing their own children inside the firm. They identify business opportunities and undertake aggressive investment strategies to prepare development path for the next generation.

In order to secure development path for the next generation, founders of family business tend to undertake higher risk when nearing their retirement. In Korea, the founder of Samsung, Byung Chul Lee decided to enter semiconductor business when nearing his retirement by acquiring Hanguk Jeonja Tongsin, a major company in telecommunications in early 1980s in spite of many scepticisms around him. He died in year 1987 at the age of 77, but since around 1995 when his son, Kun Hee Lee was the chairman of Samsung, Samsung's semiconductor business has started to be successful and now Samsung is the most competitive semiconductor and smartphones manufacturing firm in the world. In Malaysia, the most prominent example is the founder of the renowned casino and resorts Genting Group, Lim Goh Tong. He has aggressively expanded the business in cruise operation, plantation and biotechnology segments during his tenure. When he passed away at the age of 89 in year 2007, his son, Lim Kok Thay took over the chairmanship and managed to transformed the gaming business into a whole new perspective with global presence in Singapore, New York, Las Vegas and UK within 10 years' time. Another example is founder of IOI Group, Lee Shin Cheng, who started his business with property development company, took a bold step at his early 40s to venture into plantation and marked another major breakthrough at his 60s to acquire Lodders Crokklan from Unilever. Besides, he set up IOI research center to improve the breeding and seed production. His business empire was inherited by his two sons, Lee Yeow Chor and Lee Yeow Seng after his death and they have created another milestone for the company by embarking on digitalization of estate operation. Founder of YTL, Yeoh Tiong Lay

established YTL & Sons Holdings which operates as a holding company during his tenure to ensure smooth succession of family business when he passed away at age 88 in year 2017. Besides, he started to invest extensively in research and development since 1990s to ensure high performance of cement and concrete products. There are many more established family firms in Malaysia transcend on the basis of father-son succession such as KKK, Scientex and Batu Kawan.

Following prior studies in employing multidimensional measures for risk-taking (Chatterjee & Hambrick, 2011; Devers, McNamara, Wiseman, & Arrfelt, 2008; Sanders & Hambrick, 2007; Serfling, 2014), we choose R&D investment, capital expenditure and financial leverage as measures for risk-taking behaviour of CEOs. R&D project is one of the policy decisions that CEO can most exert their influence over by virtue of his/her privilege inherent in the position and can formulate according to their own risk preference (Rashad Abdel-Khalik, 2014). An example would be CEOs of GlaxoSmithKline and Pfizer terminate some R&D projects and close R&D facilities after spending hundreds of millions of dollars on these projects. R&D and capital expenditure are risky investment as they involve high cost in the year they are incurred, and may not lead to revenue for several years in the near future (Cazier, 2011; Lu & Wang, 2018). However, when these investments lead to enhancement of technology, improvement in process and introduction of superior products, it results a higher level of income and growth for a firm (Trajtenberg, 1990). Leverage is an important and flexible source of external finance for growth of the firm even though it may increase the firm risk (Norden & van Kampen, 2013). Studies have shown that leverage does not affect the growth of firms with good investment opportunities (Lang, Ofek, & Stulz, 1996).

This paper offers several contributions. First, we explore the potential existence of nonlinear nexus between CEO age and managerial risk taking. Researchers may have long been aware of different risk attitude of individual at various life cycle, but not much studies have been done on risk behaviour of CEOs throughout their career life. Furthermore, even though extensive research has been conducted on the impact of CEO age on firm risk-taking, but most of this research investigate the linear relationship. To the best of our knowledge, prior studies have yet to examine the nonlinear function in the context of CEO. Therefore, by using the career life of CEO to investigate the nonlinear relationship between CEO age and risk-taking of the firm, it enhances our understanding of risk-taking behaviour of CEO. We exhibit the existence of inverse U-shaped relationship using career concern and career horizon effect, and the possible inverse S-curvilinear relationship with the presence of family CEO due to socioemotional wealth.

Second, our findings can be generalised to family business in Asian countries due to similar family dynastic succession setting. Different from western countries in which most of the family successors only hold a marginal ownership and control upon the retirement of the founder (Burkart, Panunzi, & Shleifer, 2003), Asian countries emphasize on transition of stock equity and management control in the hand of family heirs (Weng & Chi, 2019; Yabushita & Suehiro, 2014). This study provides evidence on how different dimensions of risk behaviour take priorities in various stages of a family CEO's life and demonstrates how father-entrepreneur undertake risk when nearing their retirement to ensure family continuity. Therefore, this study may provide some insights on risk-taking behaviour and course of action that can be utilised by

family business to enhance perpetuation and secure continuity of family business in Asian countries.

Third, this study also contributes to empirical research on emerging economies which lag behind developed countries in innovation and entrepreneurial initiatives. The properties of old-aged family CEOs have an important impact on risky activities. We provide evidence that family perpetuation goal can alleviate risk averse behaviour of old-aged CEOs and promote strategic risky investment. This practical implication provides insights to family owners and policymakers who aspire to promote investment activities for long-term firm survival and economic growth of the country (Memili, Fang, Chrisman, & De Massis, 2015).

Lastly, we demonstrate that observable characteristics of CEOs can be used directly to deduce the strategic decision of the firm, as suggested by upper echelons theory. Investors who takes ownership interest of a firm can diversify risk through ownership portfolio may prefer firm with riskier corporate policies. Thus, CEO age may serve as one of the indicators for investors in selecting their securities portfolios. Besides, it allows a firm to match its preferred risk strategies by considering CEO age when making hiring decision.

Theory and hypotheses

Managerial risk choice is critical in decision making as it affects economic performance, growth and survival of an organisation (Bromiley, 1991). Managerial risk-taking improves the competitive advantage of an organisation (Hoskisson, Chirico, Zyung, & Gambeta, 2017). According to Palmer and Wiseman (1999), managerial risk-taking is different from organisational risk even though organisational risk is influenced by managerial risk-taking. Their test of discriminant validity shows that both risks are not isomorphic as top managers, namely CEOs may choose strategy that align their organisation with environmental conditions. CEO is the central decision maker of a firm (Zahra & Pearce, 1989). CEOs have major risk taking responsibilities and their careers rely on how well they deal with risk (MacCrimmon & Wehrung, 1990). By virtue of power inherent in the position, CEO is presumed to have more actual influence to policy decision than any other individual in the organisation (Boivie, Lange, McDonald, & Westphal, 2011).

Upper echelons theory suggests that organizational strategic choices are partially predicted by value and cognitive bases of the executives (Hambrick & Mason, 1984). The theory argue that characteristics of an individual affects his/her interpretation of the situation. Thus, the act of organisation such as risk-taking activities can to some extent be explained from the background characteristics of CEO, and CEO age is a source of innate risk preference that affects the risky behaviour of the organisation (Serfling, 2014).

Age determines the ability of processing information and cognitive ability of a person. Older managers have lesser ability to integrate information base effectively in making decision (Taylor, 1975) and to evaluate and manage risk appropriately (Yao, Sharpe, & Wang, 2011). However, research on development of wisdoms suggest that older adults are better in using immediate recognition, intuition, guessing and inference prior to make judgement (Pliske & Mutter, 1996). Age is related to the career life of a

CEO with different risk-taking tendency at different stage of career life. At the beginning of their career life, younger managers have stronger incentive to undertake more aggressive strategy to signal their abilities due to career concern (Li et al., 2017; Prendergast & Stole, 1996). Fama (1980) is the seminal work that posits the career concern of a young manager who faces greater market scrutiny as she/he does not have reputation as high quality manager. Managers who perform poorly in the early of their career life can significantly reduce their future career opportunities (Serfling, 2014). As a result, managers in their early career life are more willing to pursue acquisition (Yim, 2013), spend on R&D (Serfling, 2014) and undertake restructuring activities (Li et al., 2017) to signal their ability.

However, beyond certain age threshold, the risk-taking behaviour is likely to be influenced more significantly by another force, which is career horizon effect. As CEOs near retirement, their career horizon shortened and become less inclined to make risky decision. This is known as CEO career horizon problem (Strike et al., 2015). Horizon problem becomes more acute as the CEO's departure date is closer (Dechow & Sloan, 1991). When nearing retirement, CEOs tend to avoid risky projects to safeguard their value of pensions (Sundaram & Yermack, 2007). Older CEOs tend to adopt less risky strategies if the outcome is only materialised after the retirement of CEOs (McClelland et al., 2012).

Shorter career horizon induces CEOs to reduce long term risky investment as they may not have sufficient time to reverse any adverse effect of the long term project (Gao, 2010). Near retirement CEOs tend to forgo breakthrough innovation as it may affect the firm performance and, thus endangering his/her job prospect and legacy in the short term (Cho & Kim, 2017). CEOs avoid risky international acquisition decisions when they approach retirement as risky strategic choices may destroy their personal reputation in their last years of employment (Matta & Beamish, 2008). Besides, CEOs nearing retirement are also negatively related to firms' long-term commitment to corporate social responsibility (CSR) as they intend to enhance their value in labour market by boosting financial performance and reducing CSR. However, this negative effect is weakened if CEOs retire at an older age or are retained on the board (Kang, 2015).

As illustrated in Fig. 1, although CEOs in the early stage of career are subject to reputation which incentivise them to take riskier investment policy, these incentives are expected to disappear when approaching retirement (Cazier, 2011). Having greater career concern to achieve success induces the younger CEO to take more risk at the beginning of career life. However, beyond certain age threshold, the risk-taking behaviour is likely to be influenced more significantly by another force, which is career horizon effect. This effect suggests that CEOs may become more conservative in their decision making when nearing retirement. As shown in Fig. 1, to the right of the inflection point, a further increase in age may result in lower risk taking as the career horizon effect dominates the investment behaviour. In equilibrium, this study expects an inverse U-shaped relationship between CEO age and managerial risk taking.

Based on these arguments, this study predicts the following hypothesis:

- **Hypothesis 1** There is an inverse U-shaped relationship between level of risk-taking and CEO age such that the relationship is positive as age initially increases but reaches an inflection point where the relationship then becomes negative

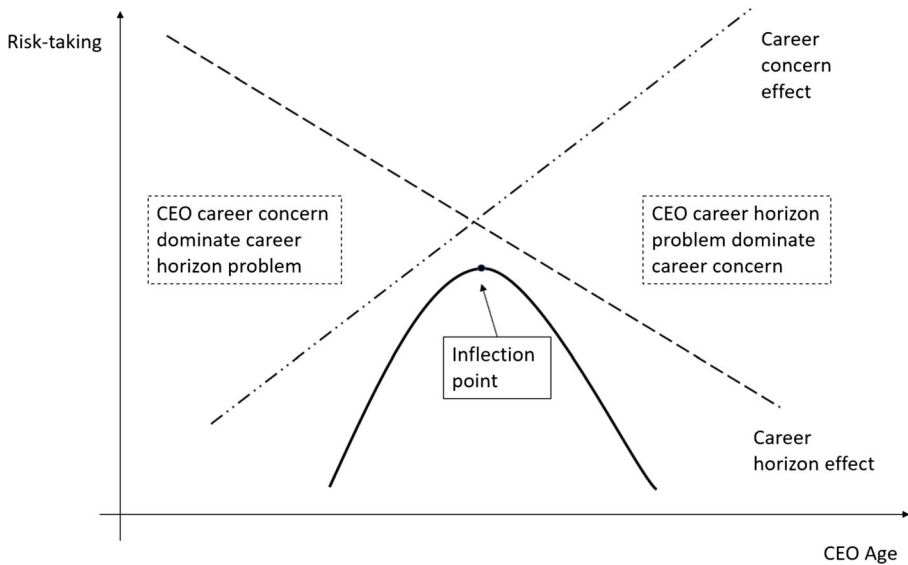


Fig. 1 Illustration of inverse U-shaped relationship between CEO age and risk-taking

The career horizon problem may be different for family CEO. According to socioemotional wealth, optimizing financial wealth may not be the ultimate objective for family business. Instead they may seek to maintain familial ownership and control for subsequent generation. (Gómez-Mejía, Haynes, Núñez-Nickel, Jacobson, & Moyano-Fuentes, 2007). They are more concern on dynastic succession and committed to preserve the perpetuation of family firms (Berrone, Cruz, & Gomez-Mejia, 2012). The near-retirement family CEOs are more likely to undertake risky but positive NPV project that may secure growth of the family business in the long run. They continue to engage in acquisition as they approach the end of their career in spite of the potential short-term risk (Strike et al., 2015). This leads to weaker horizon problem in family firms.

As stated in Fig. 1, the inverse U-shaped relationship is explained by the career concern and career horizon effect on the risk-taking tendency of a CEO. Similar to non-family CEOs, younger CEO in family firm has career concern in the beginning of their career life. CEOs in family firms may also reduce long-term projects with uncertain outcome as they do not necessarily have the time to reverse any adverse effects. However, as the CEO in a family firm approaches retirement age, a different dimension enters into play: the aspiration to preserve the ongoing renewal of family dynasty. Succession shift the CEO's attention from controlling the company to securing the continuity of family business (Strike et al., 2015). The strong concern for securing family legacy will be more pronounced for near-retirement CEOs because retirement strengthen aspiration on preserving the wealth-generating capabilities of family assets for future generation (Curasi, Price, & Arnould, 2004). Thus, the relationship between risk-taking and CEO age could be better captured using an inverse S-curve shape with the interaction of family CEO.

There will be two inflection points for age threshold which indicates the relative minimum and relative maximum of risk taking. For an inverse S-curve relationship, relative maximum risk-taking level occurs at younger age and relative minimum risk-taking

level occurs at older age. Career concern effect result a relative maximum level of risk-taking as younger family CEOs undertaking aggressive investment policies at the beginning of their career life. This is the first inflection point for the first age threshold. After this, the level of risk-taking decreases due to career horizon effect and reach a relative minimum risk-taking level. This is the second inflection point for the second age threshold. Beyond this second inflection point, the level of risk-taking increases again as the socioemotional wealth takes effect.

Therefore, the full spectrum of age-risk taking relationship is described as following an inverse S-curve. As shown in Fig. 2, the relationship between family CEO age and risk taking is positive at young age. When the age reaches the first age threshold, there is an inflection point (first inflection point) and the relationship turns negative, with level of risk-taking reducing with increases in CEO age. Again, if the CEO age goes beyond the second age threshold, another inflection (second inflection point) takes place and the relationship becomes positive, with level of risk-taking increasing with increases in CEO age.

Based on these arguments, this study predicts the following hypothesis:

- **Hypothesis 2** There is an inverse S-curvilinear relationship between level of risk-taking and CEO age if the CEO is a family CEO, such that the relationship is positive at younger age before reaching the first inflection point, negative at middle age and positive again at old age after exceeding the second inflection point

Method

Data collection

We use a sample of public companies listed on main board of Bursa Malaysia from year 2009 to 2017. Finance companies, REITS, closed-end funds and exchange traded

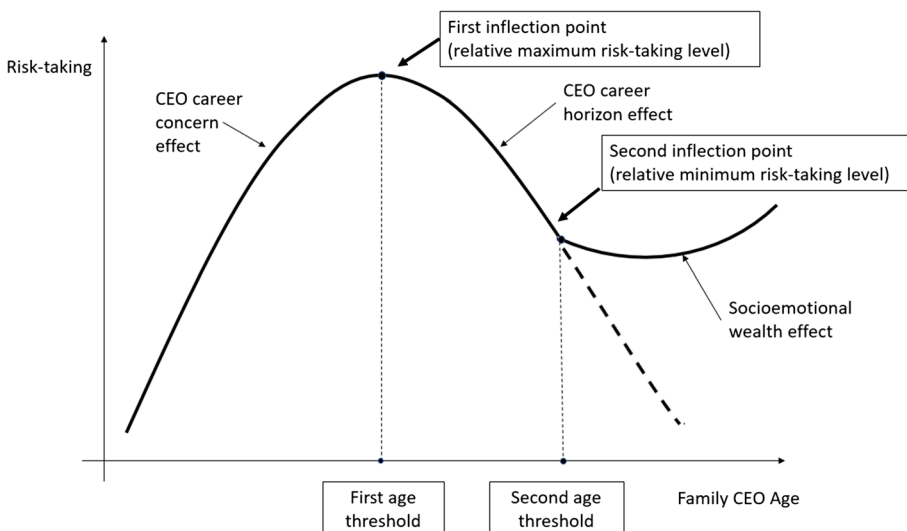


Fig. 2 Illustration of inverse S-curvilinear relationship between CEO age and managerial risk taking

funds are excluded from the sample. The background of CEOs and board characteristics are collected by hand from the profile of board of directors in the annual report, while firm financial data are obtained from Datastream. After removing firms with incomplete data on dependent and independent variables, we created a final panel consists of 6169 firm-year observation.

Measures

Dependent variable

The dependent variable in this study is R&D expenditure, capital expenditure and financial leverage. R&D expenditure (*R&D*) is measured as R&D expenses divided by book value of assets (Serfling, 2014). R&D expenditure are normally fully expensed, nevertheless we include also capitalised R&D expenditure. This is because Malaysia adopts International Financial Reporting Standard (IFRS) which allow an entity to capitalise R&D expenditures if the project is commercially viable and technically feasible.¹ Prior research suggests that capitalising R&D is appropriate as R&D investment represent future economic benefit (Han & Manry, 2004). Capital expenditure (*Capex*) is measured as capital expenditure divided by book value of assets (Coles, Daniel, & Naveen, 2006). Financial leverage (*Leverage*) is measured as long-term debts over book value of total assets (Hall, 2012).

Independent variable

Our independent variable is CEO age and we follow prior research to compute natural logarithm of CEO age (*LogCEOAge*) in the given year (Serfling, 2014).

Moderator

The moderating variable is family CEO. Family CEO (*FAM*) is a dummy variable equal to one if the CEO owns at least 5% of the shares, or has family relationship with directors or major shareholders of the family firm. Following Anderson and Reeb (2003), we define a firm as family control when one of the following criteria is met: (1) the founder or a descendant sits on the board and/or is a substantial shareholder; (2) at least two board members are related by blood or marriage. According to Section 136 of the Companies Act 2016, a substantial shareholder is defined as: “A person who has an interest in one or more voting shares in a company and the number or the aggregate number of such shares is not less than 5% of the total number of all the voting shares included in the company”.

Control variables

Based on previous studies on risk-taking (Elnahas & Kim, 2017; Habib & Hasan, 2017; Lu & Wang, 2018; Rashad Abdel-Khalik, 2014; Serfling, 2014), we control a series of

¹ <https://www.accaglobal.com/hk/en/student/exam-support-resources/fundamentals-exams-study-resources/f7/technical-articles/rd.html>

factors that may be associated with risk-taking. We control for tenure (*LogTenure*, measured as natural logarithm of no. of years the CEO has been with the firm), gender (*Gender*, measured as a dummy variable, which takes one if the CEO is female), return on asset (*ROA*, measured as ratio of earnings before interest & taxes to total sales), firm size (*LogFirmSize*, measured as natural logarithm of the book value of total assets), market-to-book (*MTBV*, measured as ratio of market value of equity to its book value), firm age (*LogFirmAge*, measured as natural log of the no. of years from the establishment of the firm to the year of observation), capital expenditure (*Capex*, measured as ratio of capital expenditures to total assets), board size (*BoardSize*, measured as total no. of directors on the board) and board independence (*IndBoard*, measured as proportion of independent directors on the board).

Estimation method

To test the relationship between CEO age and risk-taking and prove the hypotheses of the study, we employ System GMM estimator for analysis of data. Increasing research related to upper echelons theory have pointed to issues associated with endogeneity. The basic logic of upper echelons theory is that executives take action based on individual information processing of the situation. However, this may not be the case if the actions are due more to mandate (Hambrick, 2007). For instance, a new CEO tends to make changes because s/he is merely executing the desire of board who intend to make major changes. It is also possible that due to selection bias, specific types of firms hiring CEO of certain ages (Li et al., 2017). For example, if the firm prefer more restructuring, younger CEO may be selected by the firm. It could also be the case that executives are drawn to setting that suit their profiles by which executives possessing technology expertise are more likely to invest a significant amount in R&D (Hambrick, 2007).

Due to this endogeneity issue, we need to consider more advanced econometric method. The use of fixed effects method could only control for unobservable firm-specific factors but would not alleviate the endogeneity problem. Endogeneity issue advocates the use of instrumental variable technique, where the lagged dependent variables and endogenous independent variables are instrumented using an appropriate set of instrumental variables. We employ system generalised method of moments (GMM) estimator which can mitigate the distortions caused by endogeneity problem. It provides consistent parameter estimates by utilising instruments that can be obtained from the orthogonality condition that exist between the lagged values of the variables and disturbances (see Arellano & Bond, 1991). It does not depend on external exogenous instruments which may be difficult to identify in practice (Wintoki, Linck, & Netter, 2012). For GMM results to be reliable and consistent, two diagnostics have to be fulfilled. The first test is a test of first-order and second-order serial correlation. As dynamic panel GMM estimator is subject to an optimal choice of instruments where the validity of instruments depends on the absence of higher-order serial correlation in the idiosyncratic component of the error term, the residuals in first differences (AR(1)) should be correlated and there should be no serial correlation in second differences (AR(2)). The second test is a Hansen test of over-identification. The test yields a J-statistic which is distributed χ^2 under the null hypothesis of the validity of our instruments.

Analyses and results

Descriptive statistics

Table 1 presents the descriptive statistics and correlations for the variables in our analysis. We show the mean, standard deviation, maximum and minimum value of the variables. The mean value of *R&D* is 0.007, which indicates that firms in Malaysia on average spend 0.7% only of total assets on R&D investment. The mean value of *Capex* is 0.035, indicating that firms' investment in property, plant and equipment is much higher than R&D in Malaysia. The mean and maximum value for *leverage* is 0.087 and 0.591 respectively, showing that Malaysian firms are rather highly leveraged. The average CEO age is 55 years old, with oldest CEO aged 96 and youngest CEO aged 23. The average tenure of CEOs is 11 years, with longest term of office up to 46 years. Among these CEOs, 68% of them are family CEOs (4219 of 6169), displaying high level of family ownership in Malaysia. This can also be reflected in the maximum CEO tenancy of 46 years as most of the firms are managed by the family founder, their heir or family member who hold the position for long period of time.

Table 2 displays the Pearson correlation matrix of main variables. In general, the correlation figures among explanatory variables and control variables are less than 0.50, indicating no problem of multicollinearity. For *LogCEOAge*, there is no multicollinearity problem for first-order *LogCEOAge*. But when the first-, second-, and third-order *LogCEOAge* are put into the model at the same time, the correlation is much larger than the threshold of 0.5. However, this indication is not relevant, because multicollinearity bias is present only when the correlated variables are a linear value of each other. In our case, the *LogCEOAge*, square and cubic are not in a linear relationship of each other, but is the respective quadratic and cubic relationship (Wooldridge, 2001).

Table 1 Summary statistics for variable characteristics

Variables	Mean	S.D.	Min	Max
<i>R&D</i>	0.007	0.019	0	0.145
<i>Capex</i>	0.035	0.043	0	0.263
<i>Leverage</i>	0.087	0.111	0	0.591
<i>CEO Age</i>	54.737	9.131	23	96
<i>Tenure</i>	10.549	8.531	1	46
<i>Gender</i>	0.043	0.203	0	1
<i>ROA</i>	0.039	0.094	-0.506	0.386
<i>LogFirmSize</i>	12.957	1.451	10.005	17.726
<i>MTBV</i>	1.165	1.552	-0.420	14.370
<i>FirmAge</i>	25.607	16.219	1	107
<i>BoardSize</i>	7.931	2.147	3	21
<i>IndBoard</i>	0.466	0.127	0.111	0.857
<i>FAM</i>	0.684	0.465	0	1

Results

We conduct a hierarchical regression approach to test the hypotheses. The dependent variable is *R&D* in Models 1–3, *Capex* in Models 4–6 and *Leverage* in Models 7–9. All continuous variables are winsorized to reduce potential bias in the results from extreme value. We test the linear relationship in Model 1, Model 4 and Model 7 even though the focus of our study is nonlinear relationship. We introduce square term of CEO age (LogCEOAge^2) in Model 2, Model 5 and Model 8 to examine H1. In order to examine H2, we include in Model 3, Model 6 and Model 9 the interaction between CEO age and family CEO ($\text{LogCEOAge} \times \text{FAM}$), squared term of CEO age and family CEO ($\text{LogCEOAge}^2 \times \text{FAM}$) and, cubic term of CEO age and family CEO ($\text{LogCEOAge}^3 \times \text{FAM}$). Both industry and year indicators are included in all regression models. The results of Arellano-Bond tests and the Hansen test are reported. The insignificant figures (at 5%) of second-order serial correlation (AR(2)) signify that there is no autocorrelation in the first difference and the insignificant figures (at 5%) of Hansen test indicate validity of instruments.

Table 3 shows the GMM regression results that test the hypothesis in the study. Model 1, Model 4 and Model 7 present the results of regressing the linear term of CEO age on R&D, capital expenditure and leverage respectively. It can be seen that CEO age has a negative linear relationship with all the three dependent variables. H1 predicts that there is an inverse U-shaped relationship between level of risk-taking and CEO age. We add the quadratic term (square) of the CEO age and observe a significant inverse U-shaped relationship. For R&D (as shown in Model 2), the coefficient estimate on LogCEOAge is 2.130 while the coefficient estimate on LogCEOAge^2 is -0.281 . We use these estimated coefficients to calculate the inflection point and found that the turning point lies around age 44.² As shown in Model 5 and Model 8, we obtained the similar inverse U-shaped relationship for capital expenditure and financial leverage with turning point lies around age 44 and 45. This result support H1 that younger CEOs undertake more aggressive strategy to signal their abilities due to career concern in the beginning of their career life. However, beyond age threshold of 44/45, their tendency to take risk declines due to career horizon effect.

H2 tests the moderating role of family CEO on the association between CEO age and risk-taking activities. In Model 3, we add the interaction between family CEO and CEO age, and the quadratic term and cubic term of CEO age in sequence in order to test the possible linear, curvilinear and S-curvilinear relationship with R&D. As expected, the linear $\text{LogCEOAge} \times \text{FAM}$ is positively ($\beta = 28.625$), the quadratic is negatively ($\beta = -7.326$), and the cubic is positively ($\beta = 0.620$) associated with R&D expenditure. Calculations carried out on these coefficients reveal that the inflection points on the cubic function of family CEO age are approximately 36 years old (a relative maximum risk-taking level) and 73 years old (a relative minimum risk-taking level).³ Similar

² The turning points is the CEO age that optimizes the quadratic equation. It is calculated using the following formula: $\text{LogCEOAge turning point} = -(1/2) (\text{coef}(\text{LogCEOAge})/\text{coef}(\text{LogCEOAge}^2))$, where $\text{coef}()$ denotes the regression coefficients (only if statistically significant) of variables LogCEOAge and LogCEOAge^2 obtained from the estimation of a given model.

³ The turning points of a cubic function are calculated as follows: Assuming all other variables are constant and denoting LogCEOAge by x : $R\&D = 28.625x - 7.326x^2 + 0.620x^3$. The turning points are found by differentiating y ($R\&D$) with respect to x , letting $\partial y/\partial x = 0$ and solving for x . To determine whether x is a maximum or minimum turning point, calculate the value of $\partial^2 y/\partial x^2$. If $\partial^2 y/\partial x^2 > 0$, the turning point is a maxima, if $\partial^2 y/\partial x^2 < 0$, the turning point is a minima.

Table 2 Pearson correlation of explanatory variables

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1 <i>R&D</i>	1														
2 <i>Capex</i>	0.023	1													
3 <i>Leverage</i>	-0.001	0.114*	1												
4 <i>LogCEOAge</i>	-0.074*	-0.022	0.002	1											
5 <i>LogCEOAge</i> ²	-0.073*	-0.024	0.000	0.998*	1										
6 <i>LogCEOAge</i> ³	-0.073*	-0.026*	-0.002	0.998*	0.999*	1									
7 <i>LogTenure</i>	-0.009	-0.023	-0.052*	0.417*	0.418*	0.419*	1								
8 <i>Gender</i>	-0.016	-0.007	0.000	-0.119*	-0.118*	-0.116*	-0.035*	1							
9 <i>ROA</i>	0.076*	0.147*	-0.035*	-0.005	-0.005	-0.006	0.061*	-0.058*	1						
10 <i>LogFirmSize</i>	-0.067*	0.122*	0.425*	0.090*	0.090*	0.090*	-0.002	-0.059*	0.191*	1					
11 <i>MTBV</i>	0.073*	0.158*	0.059*	-0.048*	-0.050*	-0.051*	-0.116*	-0.026*	0.356*	0.133*	1				
12 <i>LogFirmAge</i>	-0.123*	-0.098*	0.044*	0.096*	0.100*	0.103*	0.087*	0.013	0.003	0.291*	0.027*	1			
13 <i>BoardSize</i>	-0.038	0.102*	0.148*	0.052*	0.053*	0.053*	-0.126*	-0.055*	0.051*	0.339*	0.114*	0.046*	1		
14 <i>IndBoard</i>	0.036	-0.067*	0.001	-0.070*	-0.072*	-0.074*	-0.067*	0.018	-0.088*	-0.055*	-0.039*	0.115*	-0.254*	1	
15 <i>FAM</i>	-0.013	-0.045*	-0.098*	0.056*	0.062*	0.0681*	0.407*	-0.008	0.019	-0.207*	-0.150*	-0.155*	-0.151*	-0.069*	1

The table reports correlations of the variables used in multivariate analysis, * denotes significance at 5% levels (two-tailed).

Table 3 GMM regression results for testing hypotheses

Variables	Dependent variable: <i>R&D</i>			Dependent variable: <i>Capex</i>			Dependent variable: <i>Leverage</i>		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9
<i>Lagged</i>	0.508*** (0.000)	0.505*** (0.000)	0.299*** (0.000)	0.318*** (0.001)	0.270*** (0.000)	0.270*** (0.000)	0.670*** (0.000)	0.5782*** (0.000)	0.6108*** (0.000)
<i>LogTenure</i>	0.004** (0.031)	0.009*** (0.007)	0.003 (0.128)	0.002 (0.573)	0.000 (0.949)	0.002 (0.138)	0.018*** (0.002)	0.000 (0.993)	0.006 (0.229)
<i>Gender</i>	0.005 (0.515)	0.013 (0.114)	-0.012* (0.052)	0.037 (0.209)	0.002 (0.881)	0.017*** (0.003)	-0.001 (0.910)	-0.038 (0.632)	-0.011 (0.599)
<i>ROA</i>	0.038*** (0.000)	0.033 (0.308)	0.005 (0.341)	0.064* (0.089)	-0.015 (0.397)	0.004 (0.540)	-0.025 (0.440)	-0.015 (0.860)	-0.053 (0.194)
<i>LogFirmSize</i>	0.001 (0.508)	0.009* (0.058)	-0.002 (0.160)	0.004 (0.441)	0.004* (0.088)	0.004*** (0.002)	0.016** (0.014)	0.038*** (0.004)	0.011* (0.065)
<i>MTBV</i>	-0.001 (0.212)	-0.002 (0.187)	-0.001** (0.028)	0.002 (0.499)	0.002 (0.211)	0.000 (0.473)	-0.001 (0.306)	-0.006 (0.436)	0.000 (0.876)
<i>LogFirmAge</i>	0.002 (0.521)	0.002 (0.695)	-0.008*** (0.003)	-0.024* (0.051)	-0.007 (0.166)	-0.005* (0.053)	0.011* (0.060)	-0.028 (0.386)	-0.014 (0.136)
<i>Capex</i>	-0.028** (0.037)	-0.074 (0.182)	-0.025 (0.130)				0.210*** (0.000)	1.359*** (0.000)	0.325*** (0.000)
<i>BoardSize</i>	0.000 (0.769)	0.000 (0.937)	-0.002*** (0.000)	0.000 (0.898)	0.002** (0.021)	0.000 (0.970)	0.003** (0.022)	-0.003 (0.521)	-0.002 (0.254)
<i>IndBoard</i>	-0.007 (0.442)	0.008 (0.677)	0.031*** (0.000)	-0.082* (0.084)	-0.092*** (0.004)	-0.009 (0.269)	0.173*** (0.003)	0.011 (0.918)	0.017 (0.570)
<i>LogCEOAge</i>	-0.033** (0.021)	2.130** (0.011)	0.911** (0.017)	-0.067** (0.047)	2.159*** (0.006)	-7.352 (0.448)	-0.047*** (0.050)	3.827** (0.031)	-62.333 (0.203)
<i>LogCEOAge²</i>		-0.281*** (0.010)	-0.501*** (0.009)		-0.283*** (0.006)	2.035 (0.419)	-0.047*** (0.028)	-0.507** (0.028)	15.892 (0.209)
<i>FAM</i>			-36.998*** (0.009)			-22.208 (0.419)			-121.725* (0.209)

Table 3 (continued)

Variables	Dependent variable: <i>R&D</i>			Dependent variable: <i>Capex</i>			Dependent variable: <i>Leverage</i>		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9
<i>LogCEOAge</i> × <i>FAM</i>			(0.000) 28.625*** (0.000)			(0.101) 17.846* (0.092)			(0.084) 94.731* (0.083)
<i>LogCEOAge</i> ² × <i>FAM</i>			-7.326*** (0.001)			-4.770* (0.084)			-24.485* (0.082)
<i>LogCEOAge</i> ³ × <i>FAM</i>			0.620*** (0.001)			0.424* (0.077)			2.102* (0.082)
<i>Industry fixed effects</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Year fixed effects</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	1570	1570	1570	5237	5237	5237	5123	5123	5123
Number of groups	226	226	226	765	765	765	765	765	765
AR(1) test	-2.472* [0.013]	-2.557* [0.011]	-2.513* [0.012]	-5.847* [0.000]	-10.454* [0.000]	-10.301* [0.000]	-8.523* [0.000]	-6.647* [0.000]	-8.348* [0.000]
AR(2) test	-1.542 [0.123]	-1.641 [0.101]	-1.856 [0.063]	0.504 [0.614]	0.837 [0.403]	0.586 [0.558]	-0.766 [0.444]	-1.453 [0.146]	-0.797 [0.425]
Hansen test	31.22 [0.651]	15.874 [0.462]	67.228 [0.216]	49.853 [0.845]	54.785 [0.521]	114.355 [0.395]	54.719 [0.090]	48.029 [0.943]	64.684 [0.488]
Inflection points (years old)	44	44	36 (relative maximum) 73 (relative minimum)	45	45	35 (relative maximum) 51 (relative minimum)	44	44	38 (relative maximum) 61 (relative minimum)

Figures in parentheses are the values of *p* value; ***, **, and * denote the significance at the 1%, 5% and 10% respectively. Figures in [] are the *p* value; * denote the significance at 5%.

results are also demonstrated for capital expenditure and leverage with positive linear (Model 6: $\beta = 17.846$; Model 9: $\beta = 94.731$), negative quadratic (Model 6: $\beta = -4.770$; Model 9: $\beta = -24.485$) and positive cubic (Model 6: $\beta = 0.424$; Model 9: $\beta = 2.102$) relationship when interaction term of family CEO is added to the regression. The relative maximum risk-taking of CEOs occurs at age 35 and the relative minimum risk-taking occurs at age 51 for capital expenditure; and age 38 and 61 for financial leverage. Among these three risky corporate policies, the relative minimum risk-taking for R&D occurs at the older age of 73. This may because as compared to capital expenditure and leverage, R&D is a long-term investment with higher level of uncertainty on future economic benefit (Bhagat & Welch, 1995).

Discussion

Our findings speak to the importance of examining the nonlinear relation of CEO age and risk-taking of the firm. We argue that age is related to career life of a CEO with different risk-taking tendency at different career stage. Consistent with managerial signalling hypothesis (Prendergast & Stole, 1996), we found that CEOs in their early career stage tend to undertake higher level of risk-taking. However, beyond their mid-40s, career horizon problem prevails (Cazier, 2011) and they begin to decrease risk-taking activities. This horizon problem may become more acute as the CEOs retirement date get closer as they intend to safeguard their value of deferred compensation (Sundaram & Yermack, 2007).

Building on this career life cycle, we further investigate if the presence of family CEO can alleviate the negative impact of career horizon on risk-taking propensity. We draw on socioemotional wealth effect (Gómez-Mejía et al., 2007) and contend that near-retirement family CEO may pursue long-term risky strategy that expand family dynasty to secure firm growth for incumbent generation. We find that the relationship between CEO age and firm risk-taking is captured using an inverse S-curve in that either a young- or old-aged family CEO will increase, while a middle-aged family CEO will reduce risk-taking activities. The relative maximum risk-taking occurs at their late-30s and relative minimum risk-taking occurs averagely at their 60s. These description of inverse S-curve relationship can be explained as follow: (1) when the family CEOs are at younger age, similar to normal CEOs, they tend to undertake more risk-taking activities as they are subject to reputation cost (Cazier, 2011). Then (2) family CEOs begin to reduce risky investment when they nearly reach 40s as they concern that they may not have the time to reverse any adverse effect in case of the failure of the risky project due to shorter horizon (Gao, 2010). Finally (3) as the family CEOs reach their 60s, motivated by socioemotional wealth effect, they undertake more risky investment policies to provide better growth opportunities for their future generation (Strike et al., 2015).

Contribution

This study has enhanced our understanding on the relationship between CEO age and corporate risk-taking. First, prior studies intensively discuss the classical linear relationship, with inconclusive findings (Bertrand & Schoar, 2003; Simsek, 2007). Some

research suggests positive and some proposed negative relationship. Our study provides comprehensive framework and consolidate prior studies on career concern, career horizon and socioemotional wealth effect to extend our understanding on the relationship between the age of CEO and the firm risk-taking strategies. Generally, we show that higher level of risk-taking may occur at several stages throughout the career life cycle of the CEO and the relationship between CEO age and risk-taking are better captured by nonlinear function. Specifically, we demonstrate the relationship between CEO age and risk-taking of family business. The desire of family CEOs or founders to preserve the family legacy motivates them to pursue risky strategic decision when nearing retirement, and as a result, inverse S-curve relationship is found in this context. As most of the firms in Asian countries are mainly family controlled, and given the similarity in institutional features of ownership structure and family dynastic succession setting between Malaysian and Asian corporations, our results can be generalised to family business in Asia.

Second, this study contributes to the literature that employ upper echelons theory in predicting firm behaviours using the characteristics of CEOs. In broader context, this study provides empirical evidence on how managerial characteristics affect the corporate risk-taking. Prior studies have focused on the role of gender, education level and working experience of CEOs in determining the riskiness of firm policy (Barker III & Mueller, 2002; Elsaid & Ursel, 2011; Hambrick & Mason, 1984). Specifically, this study adds to the growing body of literature exploring implication of CEO age to risky investment policy (Li et al., 2017; Serfling, 2014; Yim, 2013).

Third, this study also offers important practical implications. We suggest that investors should realise that not all old-aged CEOs have lower propensity to take risk. Our work enriches the accepted notion that CEOs nearing retirement undertake lower level of risky investment due to shorter career horizon. We demonstrate that socioemotional aspect of family-owned firm can act as a buffer for the conservative behaviour of older CEOs. These findings facilitate investors to select their securities investment portfolio by using the observable characteristics, namely the age of the CEO and his/her ownership of the firm.

Limitation and future research

This study contains certain caveats that suggest for avenue for future research. First, although we have generally argued that the inverse U-shaped and inverse S-curvilinear relationship can be generalised to family business in Asian countries due to similar concern on transgenerational control and family legacy, we still know little concerning how family firms may behave if they are in their second or third generation due to longer family history. Thus, it opens interesting avenue to test this nonlinear relationship in family firms in other Asian countries.

Second, previous work emphasize that even low level of endogeneity is a critical problem for research practice as it produces biased and inaccurate result (Semadeni, Withers, & Trevis Certo, 2014). There are several approaches to address this endogeneity issue but this study only employs GMM estimator for analysis of the data. Other than GMM techniques, there are other approaches to deal with endogeneity issues such as Heckman two-step procedure and propensity score matching which are commonly used in age-risk taking research (Li et al., 2017; Serfling, 2014). While in

some cases, certain techniques might be equally suitable to implement, but there are several factors such as research design, data collection instruments and complexity of the model that future researchers can take into consideration when selecting the most suitable treatment.

Availability of data and material Not applicable.

Code availability Not applicable.

Authors' contributions Not applicable.

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Compliance with ethical standards

Conflicts of interest/competing interests Not applicable.

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